PROPOSED CHANGE

Break SABRE ties by two-qubit depth

Among options SABRE already scores as equal, prefer the one that adds the least two-qubit depth.

A tie-break only. It never changes which trials or SWAPs are considered, or any heuristic score — so the number of SWAPs, and the two-qubit gate count, stay the same. Only depth drops.

layout-trial key: $(\text{swaps}, \text{index}) \rightarrow (\text{swaps}, \text{est. 2q depth}, \text{index})$

On by default. 3 source files, +159 / −6. crates/transpiler/src/passes/sabre/{dag,layout,route}.rs

BACKGROUND

SABRE routing in one loop

π: logical→physical layout · a two-qubit gate is routable when its operands sit on adjacent physical qubits.

1

Front layer

Gates whose dependencies are met. Route the ones already adjacent.

→

2

Candidate SWAPs

Hardware edges touching a qubit in a blocked front-layer gate.

→

3

Score each SWAP

Distance to the targets, lookahead, optional decay.

→

4

Apply & repeat

Insert the best SWAP, update π, route newly adjacent gates.

Layout search runs this in forward/reverse passes over several trials to pick an initial layout; routing keeps the trial with the fewest SWAPs. Two places in this loop routinely produce **exact ties** — next slide.

THE GAP

Today, ties break arbitrarily

SABRE's scoring often rates several options exactly equal. It then picks with no regard to the depth of the result.

- **Layout-trial selection.** Several trials finish with the **same swap count**. SABRE keeps the one with the lowest trial index — `min(swap_count, index)`. Index has nothing to do with circuit quality.
- **SWAP selection.** During routing, several candidate SWAPs share the **best heuristic score**. SABRE picks one of them **uniformly at random**.

> Both tie-breaks ignore the two-qubit depth the choice produces. Among options that are otherwise equal, one often routes the circuit's gates in fewer layers — that depth is left on the table by a coin flip.

THE CHANGE

Resolve the tie by two-qubit depth

Track, per physical qubit, the running two-qubit depth reached so far.

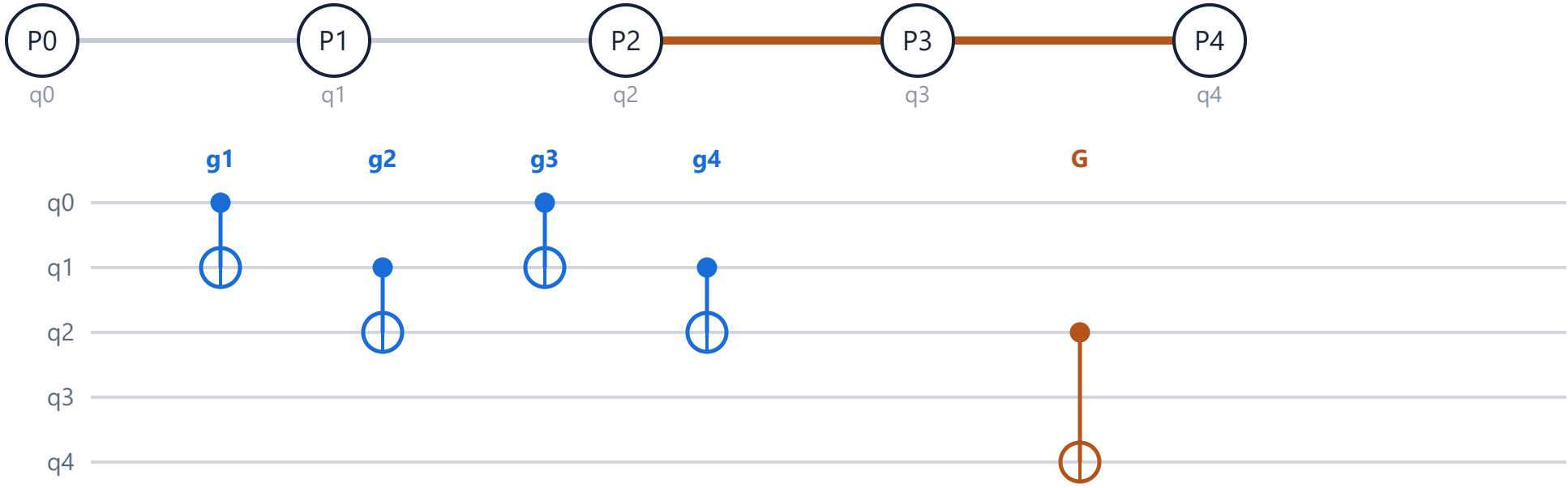
a routed 2q node on (a, b): $\mathrm{depth}[a] = \mathrm{depth}[b] = \max(\mathrm{depth}[a], \mathrm{depth}[b]) + (\text{its 2q-gate count})$

- **SWAP tie-break** (`route.rs`): among the SWAPs tied for the best heuristic score, keep those whose **deeper qubit ends up shallowest**, then fall back to the existing random choice. The tracking is rolled back with SABRE's release valve. (`dag.rs` stores each node's 2q-gate count, `num_2q` .)
- **Layout-trial key** (`layout.rs`): `(swap_count, estimated 2q depth, index)` . The estimate replays the routed order once, counting a SWAP as three two-qubit gates.

> It only re-orders options already judged equal on swap cost, so the number of SWAPs — and therefore the **two-qubit gate count — cannot change**. The routed gates are the same; they are packed into fewer layers.

One busy chain, two idle wires

5-qubit line. A prefix of gates keeps P0–P2 busy; P3 and P4 sit idle. Then a gate G = CX(q2, q4) must route.



G's qubits q2 and q4 are 2 edges apart, so one SWAP is needed. The blue prefix shares q1, so those gates stack up on P0–P2; P3 and P4 never get used.

Both SWAPs fix G — the heuristic ties

Per-qubit two-qubit depth after the prefix (each CX adds one layer to both its qubits):

physical wire	P0	P1	P2	P3	P4
2q depth so far	3	4	4	0	0

A: SWAP(P2, P3)

Brings q2 next to q4. Lands on **P2**, which is already at depth 4. Deeper endpoint depth = **4**.

B: SWAP(P3, P4)

Also brings the pair adjacent. Lands on **idle** P3–P4. Deeper endpoint depth = **0**.

> Both SWAPs reduce G's distance from 2 to 1, so SABRE's distance/lookahead score rates them **exactly equal**. Today it would pick between A and B at random.

The tie-break keeps the shallower route

quantity	A: SWAP(P2,P3)	B: SWAP(P3,P4)
heuristic score (distance)	−1	−1 (tie)
SWAPs inserted	1	1 (same)
deeper endpoint's 2q depth	**4**	**0**
finish depth of the block	**8**	**5**

Same heuristic score, same one SWAP, same two-qubit gate count. But A stacks the SWAP on the already-deep P2 chain (4 + 3 + 1 = 8 layers), while B runs it on the idle wires and overlaps the busy prefix (max(4,3) + 1 = 5).

> **Without the tie-break:** a 50/50 coin flip between A and B. **With it:** always B — the shallower route. Same gates, fewer layers. Applied across every tie in layout search and routing, this is where the two-qubit-depth reduction comes from.